

Kafka Lab : Kafka Producer Execution

Duration : 60 Min.

Scenario

Publish structural events feed into a Kafka topic.

Input Details

The environment has been pre-configured with the following resources: - **Kafka Broker:** Pre-configured on localhost:9092

Username & Naming Conventions

Your candidate prefix is a combination of your username (the part of your email/login before the @ or _ symbol) and exam code. For example, if your username is student (or student_labskraft.com) and your exam code is exam123, your prefix is **student-exam123**.

You must name your resources accordingly:

- **Topic Name:** events-stream-<prefix> (e.g. events-stream-student-exam123)

Task Objectives

1. Build Producer Script

- Publish formatted JSON alerts payload to the topic.

Verification

Once you have performed the tasks, you can run the verification script to check your progress and receive your score. The verification system will check the actual resources in the environment.

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	JSON structure verification	4 Marks
TC2	Message ingestion count check	4 Marks
TC3	Broker connectivity check	4 Marks

TC4	Reserved validation	4 Marks
TC5	Reserved validation	4 Marks

Total Score: 20 Marks

Important Notes

- Ensure all resource names contain your candidate username prefix in lowercase.
- Check that your script compiles cleanly and contains no syntax errors.